CONTROL AUTHORITY BOUNDARIES: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

This paper is not submission-ready for ICLR main. The original research bet was that a robot should maintain an explicit boundary over whether authority belongs to the learned policy, the low-level controller, or a human operator. We rebuilt the repository around a deterministic closed-loop shared-autonomy benchmark with four tasks, five distribution shifts, seven seeds, nine authority-allocation methods, seven ablations, and a stress sweep. The evidence is negative. On the combined authority-stress split, the proposed boundary reaches 0.516 task success, while a control-barrier-function safety filter reaches 0.980 and MPC risk arbitration reaches 0.976. The paired task-success difference against the strongest non-oracle baseline is -0.464 with a 95 percent CI of 0.176. Core ablations also contradict the mechanism: a safety-only boundary reaches 0.978 success and a minus-intent ablation reaches 0.946 success. The honest terminal decision is **KILL/ARCHIVE**.

1 QUESTION

The draft asked whether shared autonomy improves when the system explicitly represents control-authority boundaries: when should a learned policy act, when should a low-level controller take over, and when should a human operator receive authority? This is a crowded area. Bayesian authority allocation, confidence-based shared control, haptic shared control, control barrier functions, nonlinear MPC, controller fusion, and seamless autonomy transitions already cover much of the obvious territory.

The rebuild therefore used a hostile gate: the proposed method must beat strong non-oracle shared-autonomy baselines under hidden physical modes, human-delay shifts, and contact-mode shifts. A merely plausible arbitration story is not enough.

2 BENCHMARK

Each episode simulates a closed-loop shared-autonomy rollout over ten control phases. The hidden state includes obstacle hazard, contact volatility, human-intent ambiguity, human command delay, controller noise, and object fragility. The benchmark covers four tasks: fragile reaching through clutter, contact-rich door opening, delayed corridor navigation, and tool alignment. It evaluates five splits: nominal shared autonomy, intent ambiguity shift, contact mode shift, human delay shift, and combined authority stress.

The main run contains 80,640 episode-level rollouts. The ablation run contains 12,544 rollouts. The stress sweep contains 47,040 rollouts. All results use seven deterministic seeds. The evidence was re-run from source on 2026-06-15 and is regenerated by:

```
python src\run_experiment.py
```

3 COMPARED METHODS

The compared methods are fixed policy authority, fixed human authority, confidence-threshold shared control, Bayesian authority allocation, a control-barrier-function safety filter, MPC risk arbitration, controller-fusion priors, the proposed authority-boundary mechanism, and an oracle author-

Method	Success	Safety viol.	Regret	Human burden
Fixed policy	0.000	0.886	0.784	0.000
Fixed human	0.265	0.567	0.052	1.000
Confidence shared	0.299	0.574	0.092	0.899
Bayesian allocation	0.736	0.563	0.012	0.698
CBF safety filter	**0.980**	0.580	0.027	**0.051**
MPC risk arbitration	0.976	0.586	0.021	0.098
Controller fusion	0.164	0.719	0.484	0.208
Proposed boundary	0.516	0.571	0.026	0.858
Oracle boundary	0.979	**0.556**	**0.009**	0.225

Table 1: Combined authority-stress results. Lower safety violation, regret, and human burden are better. The proposed method loses decisively on task success and human burden.

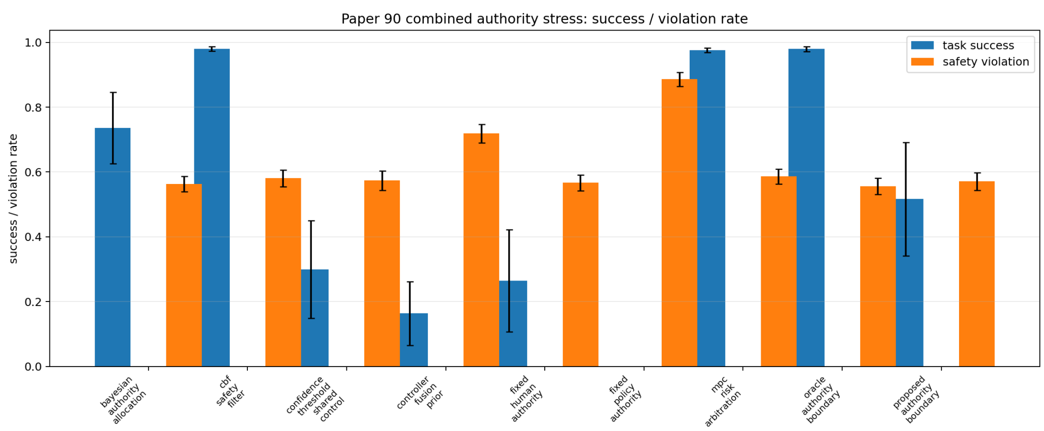

Figure 1: Success and safety-violation rates on the combined authority-stress split. The proposed boundary does not clear CBF or MPC.

ity boundary. The proposed method scores authority choices using predicted physical consequence, intent ambiguity, recovery burden, and handoff hysteresis. The oracle has access to hidden-mode utility and is included only as an upper-bound diagnostic.

4 Combined-Stress Results

Table 1 shows the main failure. The proposed boundary is safer than fixed policy authority and controller fusion, but it is not competitive with CBF or MPC. It also places a heavy load on the human operator.

The paired seed/task comparison uses CBF safety filtering as the strongest non-oracle success baseline. The proposed method's task-success difference is -0.464 with a 95 percent CI of 0.176. That is not a noisy near miss; it is a mechanism-level failure for the claimed contribution.

5 Ablations

Table 2 is more damaging than the baseline comparison. Removing intent ambiguity or using only the safety boundary improves success dramatically. A proposed mechanism whose stripped versions dominate the full version is not ready for a main-conference submission.

Ablation	Success	Safety viol.	Regret	Human burden
Full boundary	0.504	0.570	0.026	0.856
Minus consequence	0.581	0.573	0.018	0.786
Minus intent	0.946	0.566	**0.005**	0.422
Minus recovery burden	0.357	**0.536**	0.047	0.976
Minus hysteresis	0.528	0.555	0.031	0.879
Confidence only	0.264	0.588	0.135	0.882
Safety only	**0.978**	0.575	0.034	**0.049**

Table 2: Ablations on the combined authority-stress split. The safety-only and minus-intent variants beat the full mechanism, falsifying the main design claim.

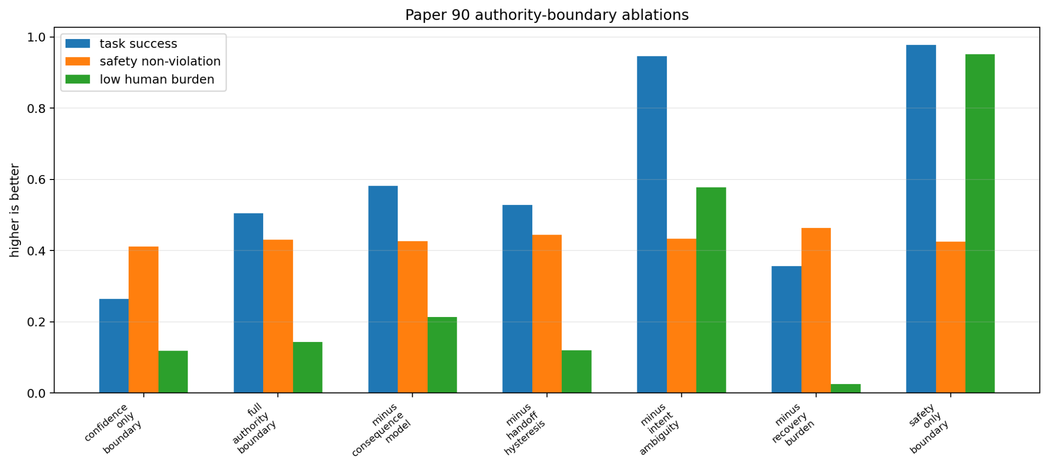

Figure 2: Ablation audit. The full authority-boundary mechanism is not the best configuration.

6 Stress Sweep

At maximum combined stress, the proposed method reaches 0.506 success. CBF reaches 0.979, MPC reaches 0.973, and Bayesian authority allocation reaches 0.742. The stress gate therefore fails. The proposed model has a useful diagnostic behavior–it avoids late handoff in many cases–but it pays for that with excessive human burden and low completion success.

7 Failure Cases

The negative cases are consistent with the quantitative result. First, under high human delay with hidden intent, the boundary over-penalizes handoff and can stay with the wrong non-human authority too long. Second, in dense fragile obstacles, the CBF-style filter reduces failures more directly than authority scoring. Third, in contact-mode shifts with low ambiguity, MPC risk arbitration solves the problem without needing an explicit authority-boundary representation.

8 Decision

The result is useful but not submission-ready. The benchmark shows that authority-boundary state is an interesting diagnostic object, but the proposed mechanism does not beat strong shared-autonomy baselines, does not survive ablation, and has no robot-hardware or high-fidelity simulator validation. The terminal recommendation is **KILL/ARCHIVE**. Revival would require a substantially different method that proves value beyond CBF, MPC, Bayesian allocation, and controller fusion on external robot evidence.

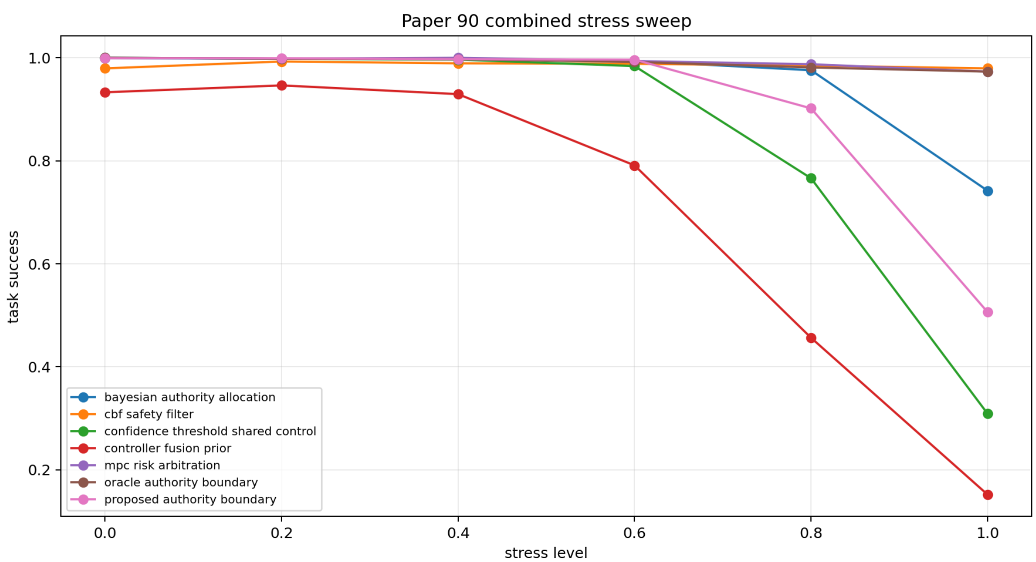

Figure 3: Combined stress sweep. The proposed boundary does not overtake CBF or MPC at high stress.